ENGR 225: Mechanics 1

Homework Coversheet

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DATE: _____

ASSIGNMENT: Homework 6

Level of involvement of AI, LLMs, or related tools in the completion of this assignment:

☐ None

■ Minor

☐ Major

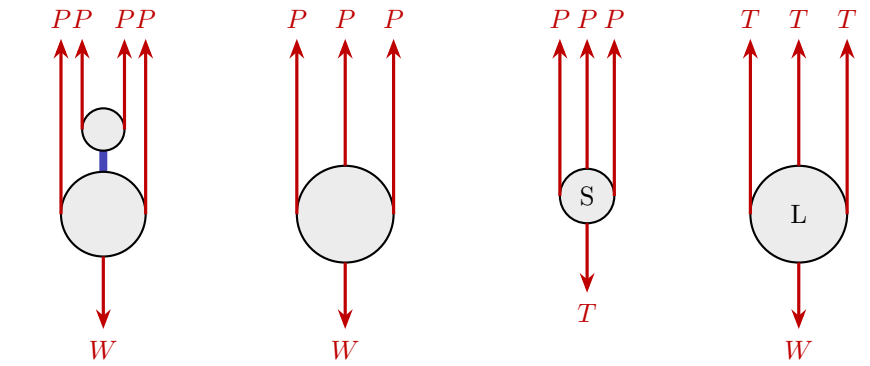
☐ Extensive

Mark the box that applies (e.g., replace ☐ with ■, or type an X beside it, before compiling/printing). Your disclosure does not affect your grade – it keeps you accountable for the work you are doing.

Problem 1

Given / Find: A 100 lb block hangs from three pulley systems. Pulleys and cords are massless and frictionless, so tension is constant along any one cord. Find the pull P for equilibrium in each.

Sketch: FBDs of the movable pulleys, cut just above them.



(a) moving block (b) lower pulley (c) pulley S (c) pulley L

Solution:

(a) Four cord segments support the block:

$$\sum F_y = 4P - W = 0 \;\Rightarrow\; P = \tfrac{100\,\text{lb}}{4} = 25.0\,\text{lb}$$

(b) Three cord segments support the lower pulley:

$$\sum F_y = 3P - W = 0 \;\Rightarrow\; P = \tfrac{100\,\text{lb}}{3} = 33.3\,\text{lb}$$

(c) Three P segments hold S; three T segments hold L.

$$\text{S:}\quad \sum F_y = 3P - T = 0 \;\Rightarrow\; T = 3P$$

$$\text{L:}\quad \sum F_y = 3T - W = 9P - W = 0 \;\Rightarrow\; P = \frac{100\,\text{lb}}{9} = 11.1\,\text{lb}$$

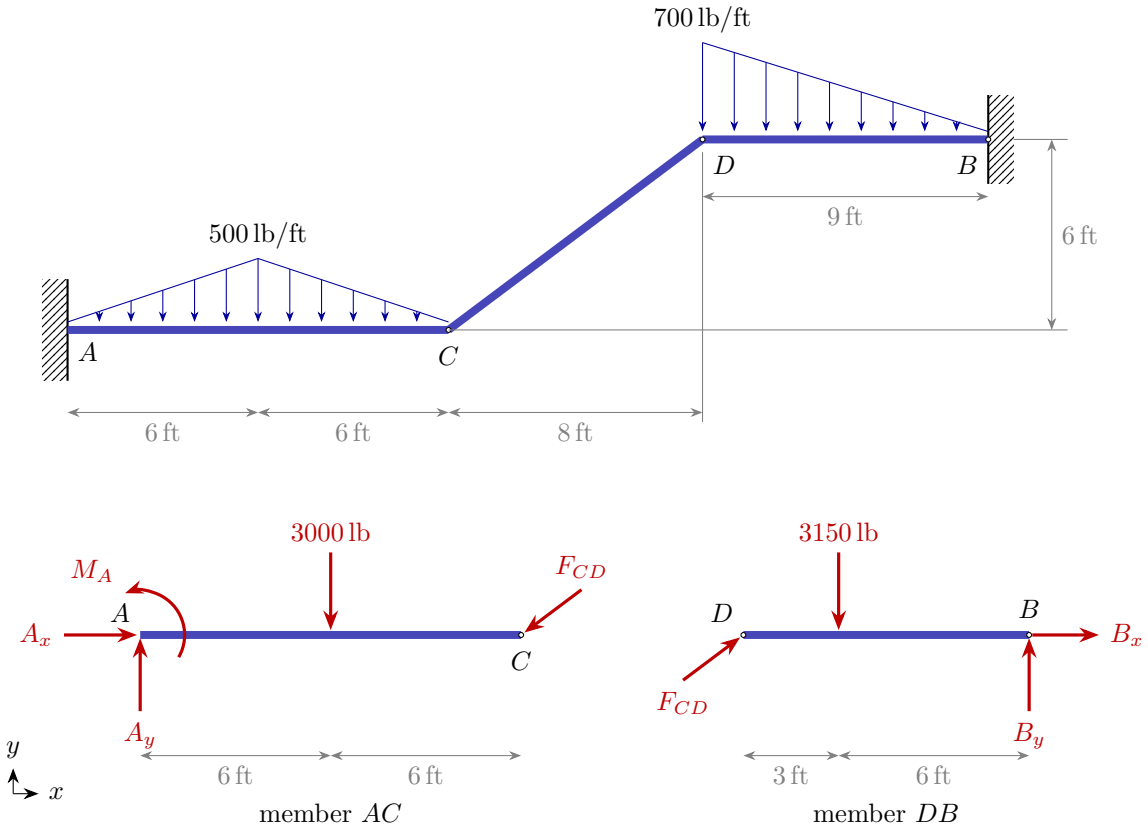
Answer

(a) $P = 25.0\,\text{lb}$ (b) $P = 33.3\,\text{lb}$ (c) $P = 11.1\,\text{lb}$

Problem 2

Given / Find: AC is fixed at A and carries a triangular load peaking at 500 lb/ft at mid-span. Link CD rises 6 ft over 8 ft. DB (9 ft) carries 700 lb/ft at D tapering to zero at the pin B. Find A_x, A_y, M_A, B_x, B_y.

Sketch:



Solution:

CD is a two-force member on a 3-4-5 slope: components $\frac{4}{5}F_{CD}$ horizontal, $\frac{3}{5}F_{CD}$ vertical (compression assumed).

Load resultants:

$$F_{AC} = \tfrac{1}{2}(500)(12) = 3000\,\text{lb at } 6\,\text{ft from } A, \qquad F_{DB} = \tfrac{1}{2}(700)(9) = 3150\,\text{lb at } 3\,\text{ft from } D$$

Member DB:

$$\sum M_B = (3150\,\text{lb})(6\,\text{ft}) - \tfrac{3}{5}F_{CD}(9\,\text{ft}) = 0 \;\Rightarrow\; F_{CD} = 3500\,\text{lb (C)}$$
$$\sum F_x = \tfrac{4}{5}(3500) + B_x = 0 \;\Rightarrow\; B_x = -2800\,\text{lb}$$
$$\sum F_y = \tfrac{3}{5}(3500) - 3150 + B_y = 0 \;\Rightarrow\; B_y = 1050\,\text{lb}$$

Member AC (CD pushes C with $(-2800, -2100)\,\text{lb}$):

$$\sum F_x = A_x - 2800 = 0 \;\Rightarrow\; A_x = 2800\,\text{lb}$$
$$\sum F_y = A_y - 3000 - 2100 = 0 \;\Rightarrow\; A_y = 5100\,\text{lb}$$
$$\sum M_A = M_A - (3000)(6) - (2100)(12) = 0 \;\Rightarrow\; M_A = 43\,200\,\text{lb ft}$$

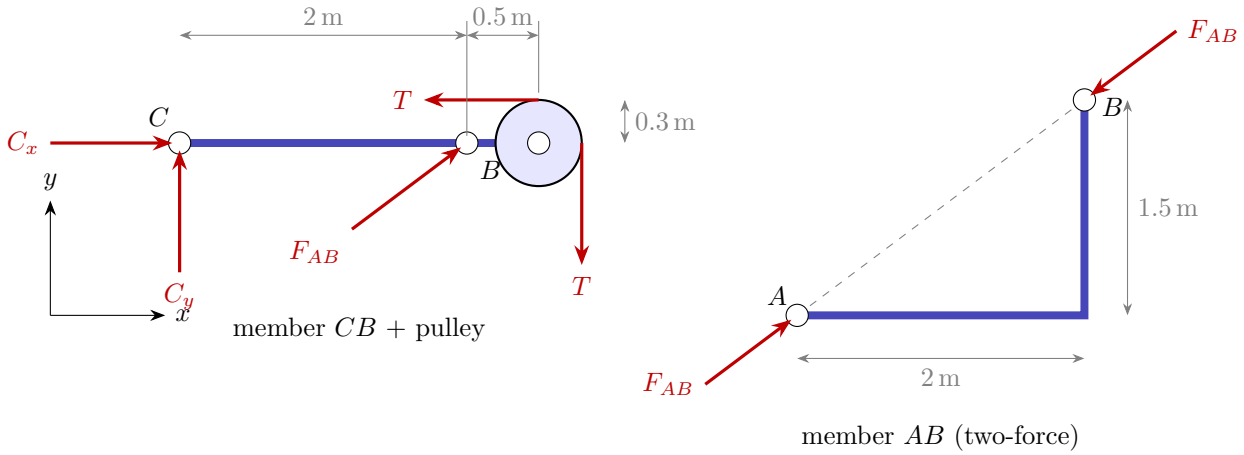
Answer

$A_x = 2800\,\text{lb}, \quad A_y = 5100\,\text{lb}, \quad M_A = 43\,200\,\text{lb ft}$ (CCW)
$B_x = -2800\,\text{lb}$ (left), $\quad B_y = 1050\,\text{lb}$

Problem 3

Given / Find: A 75 kg cylinder hangs from a cable over a 0.3 m pulley at the end of CB, then runs horizontally back to the wall. Bent member AB is pinned at A, runs 2 m right and 1.5 m up to B; the pulley center is 0.5 m past B. Find the force components at pins B and C.

Sketch:



Solution:

$T = mg = (75)(9.81) = 735.75\,\text{N}$.

AB is a two-force member along a 3-4-5 line; assume compression, so it acts on CB at B with $(\frac{4}{5}F_{AB}, \frac{3}{5}F_{AB})$.

Member CB:

$$\sum M_C = \tfrac{3}{5}F_{AB}(2\,\text{m}) + T(0.3\,\text{m}) - T(2.8\,\text{m}) = 0 \;\Rightarrow\; F_{AB} = \frac{(2.5)(735.75)}{1.2} = 1532.8\,\text{N}$$

$$B_x = \tfrac{4}{5}(1532.8) = 1226\,\text{N}, \qquad B_y = \tfrac{3}{5}(1532.8) = 919.7\,\text{N}$$

$$\sum F_x = C_x + 1226 - 735.75 = 0 \;\Rightarrow\; C_x = -490.5\,\text{N}$$

$$\sum F_y = C_y + 919.7 - 735.75 = 0 \;\Rightarrow\; C_y = -183.9\,\text{N}$$

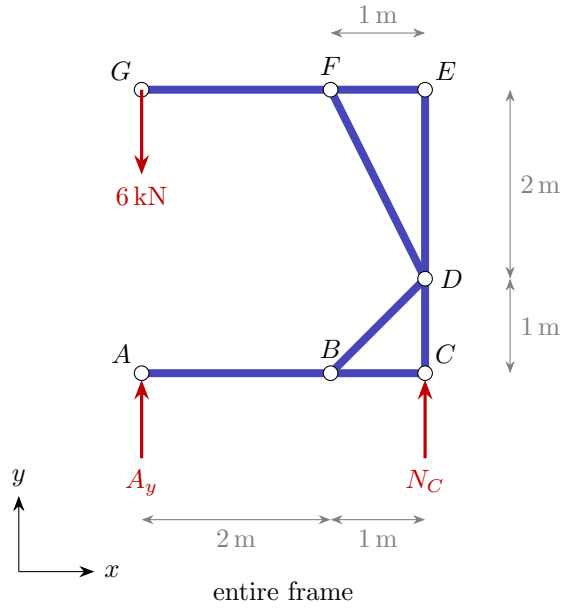
Answer

$B_x = 1230\,\text{N}$, $B_y = 920\,\text{N}$ (AB in compression)
$C_x = -491\,\text{N}$, $C_y = -184\,\text{N}$

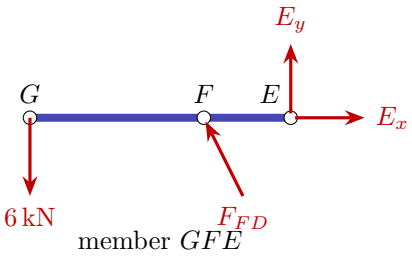
Problem 4

Given / Find: The frame rests on rollers at A and C with 6 kN down at G. Find the forces in FD and BD and the pin reactions at C on ABC and on CDE.

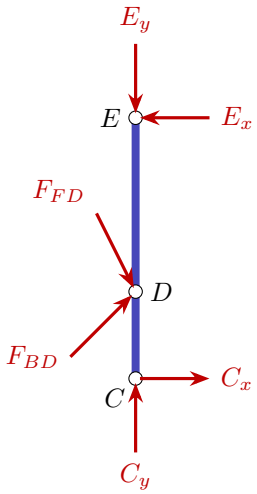
Sketch:



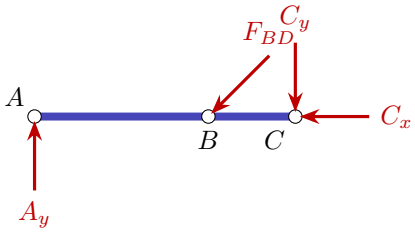
entire frame



member GFE



member CDE



member ABC

Solution:

FD (length $\sqrt{5}$ m) and BD (length $\sqrt{2}$ m) are two-force members, assumed in compression.

Entire frame: $\sum M_A = N_C(3\,\text{m}) = 0 \Rightarrow N_C = 0$, so $A_y = 6$ kN.

Member GFE:

$$\sum M_E = (6)(3) - \tfrac{2}{\sqrt{5}} F_{FD}(1) = 0 \;\Rightarrow\; F_{FD} = 9\sqrt{5} = 20.1\,\text{kN (C)}$$
$$\sum F_x = E_x - \tfrac{1}{\sqrt{5}}(20.1) = 0 \;\Rightarrow\; E_x = 9\,\text{kN}$$
$$\sum F_y = E_y + 18 - 6 = 0 \;\Rightarrow\; E_y = -12\,\text{kN}$$

Member CDE ($(-9, 12)\,\text{kN}$ at E, $(9, -18)\,\text{kN}$ from FD and $\tfrac{1}{\sqrt{2}} F_{BD}(1, 1)$ from BD, both at D):

$$\sum M_C = (9)(3) - (9)(1) - \tfrac{1}{\sqrt{2}} F_{BD}(1) = 0 \;\Rightarrow\; F_{BD} = 18\sqrt{2} = 25.5\,\text{kN (C)}$$
$$\sum F_x = -9 + 9 + 18 + C_x = 0 \;\Rightarrow\; C_x = -18\,\text{kN}$$
$$\sum F_y = 12 - 18 + 18 + C_y = 0 \;\Rightarrow\; C_y = -12\,\text{kN}$$

Answer

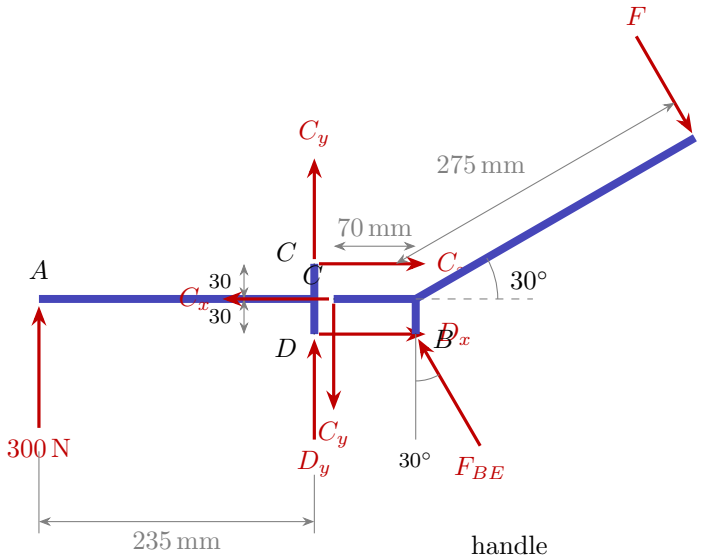
$F_{BD} = 25.5\,\text{kN (C)}, \quad F_{FD} = 20.1\,\text{kN (C)}$
Pin C on CDE: $C_x = -18.0\,\text{kN}$, $C_y = -12.0\,\text{kN}$
Pin C on ABC: $C_x = 18.0\,\text{kN}$, $C_y = 12.0\,\text{kN}$

Problem 5

Given / Find: A toggle clamp must press on the workpiece at A with 300 N. Find the handle force F, applied perpendicular to the handle.

Assumptions: C and D sit 30 mm above and below the arm centerline, 235 mm from A. B is on the centerline, 70 mm right of C. Link BE is 30° from vertical, and the handle is 30° above horizontal. As in the textbook solution, the 275 mm is measured along the handle from the point directly above B at the level of C.

Sketch:



Solution:

Clamp arm, $\sum M_D$:

$$\sum M_D = -(300)(235) - C_x(60) = 0 \;\Rightarrow\; C_x = -1175\,\text{N}$$

The arm pushes the handle at C with 1175 N to the right.

Handle. BE (two-force, compression) acts on B along $(-\sin 30°, \cos 30°)$; F acts along $(\sin 30°, -\cos 30°)$.

$$\sum F_x = 1175 - F_{BE}\sin 30° + F\sin 30° = 0 \;\Rightarrow\; F_{BE} = 2350 + F$$

$\sum M_C$, with $B = (70, -30)$ mm and F at $(70 + 275\cos 30°,\ 275\sin 30°) = (308.2,\ 137.5)$ mm:

$$\begin{aligned}\sum M_C &= F_{BE}\left[(70)\cos 30° - (30)\sin 30°\right] - F\left[(308.2)\cos 30° + (137.5)\sin 30°\right] = 0\\ &= 45.62\,(2350 + F) - 335.7\,F = 0\\ F &= \frac{(45.62)(2350)}{335.7 - 45.62} = 369.6\,\text{N}\end{aligned}$$

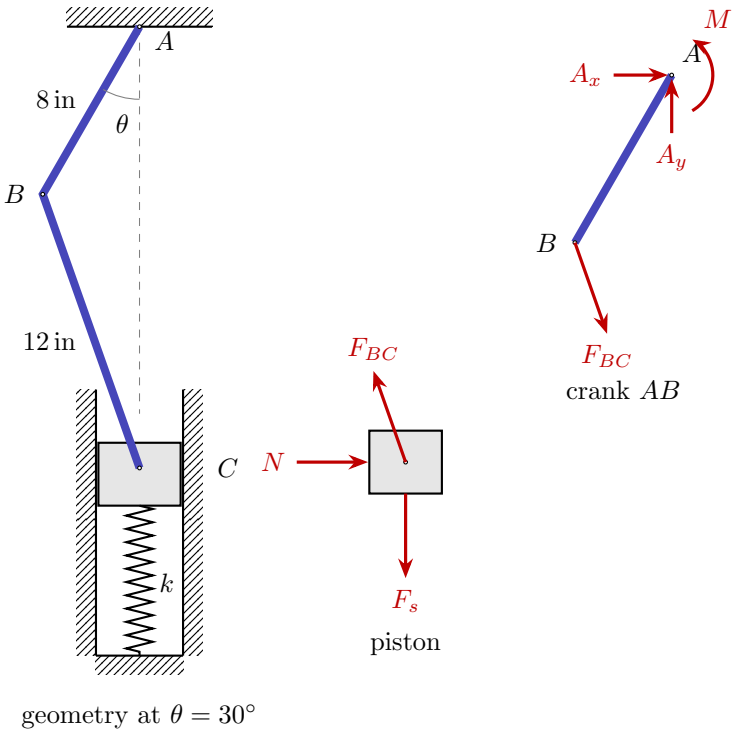
$F_{BE} = 2720\,\mathrm{N}$ (C).

Answer
$F = 370\,\mathrm{N}$

Problem 6

Given / Find: Crank AB (8 in) and link BC (12 in) drive piston C between smooth vertical walls. The spring ($k = 15$ lb/in) is unstretched at $\theta = 0°$. Find the couple M on AB for equilibrium at $\theta = 30°$.

Sketch:



geometry at $\theta = 30°$

Solution:

Geometry (origin at A). $B = (-8\sin 30°, -8\cos 30°) = (-4, -6.928)$ in. BC spans 4 in horizontally and $\sqrt{12^2 - 4^2} = 11.314$ in vertically, so C is 18.242 in below A (20 in at $\theta = 0°$):

$$\delta = 20 - 18.242 = 1.758 \text{ in}, \qquad F_s = k\delta = (15)(1.758) = 26.37 \text{ lb}$$

Piston (walls push horizontally only):

$$\sum F_y = F_{BC}\left(\tfrac{11.314}{12}\right) - 26.37 = 0 \;\Rightarrow\; F_{BC} = 27.97 \text{ lb}$$

Crank AB. BC (tension) pulls B with $F_{BC}\left(\tfrac{4}{12}, -\tfrac{11.314}{12}\right) = (9.32,\ -26.37)$ lb:

$$\sum M_A = M + \big[(-4)(-26.37) - (-6.928)(9.32)\big] = M + 170.1 = 0$$
$$M = -170.1 \text{ lb in}$$

Answer
$M = 170$ lb in (clockwise)

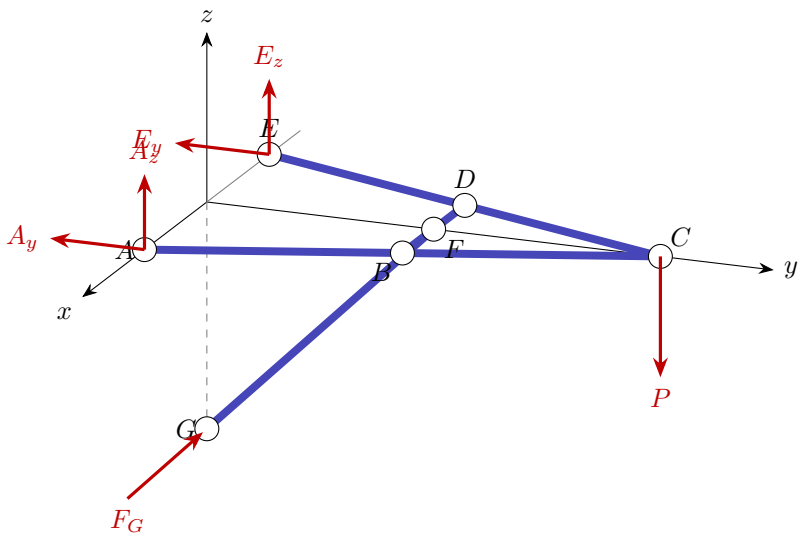
Problem 7

Given / Find: The "A" frame is held by smooth collars at A and E and a pin at G. B, C, D, F are ball-and-sockets, with F midway between B and D. The pin at G fails at 800 N. Find the largest P (load $-P\mathbf{k}$ at C) and the force components BD exerts on ABC and EDC.

Assumptions: The collars slide on a shaft along x, so they resist only y and z forces. The pin at G resists force only. Coordinates (m):

$$A(0.3,0,0),\ E(-0.3,0,0),\ G(0,0,-0.6),\ C(0,1.2,0),\ B(0.15,0.6,0),\ D(-0.15,0.6,0),\ F(0,0.6,0)$$

Sketch:



Solution:

Member FG (two-force, compression): $\vec{GF} = (0, 0.6, 0.6)$ m, so the force on the frame at G is $F_G\,\mathbf{u}_{GF}$ with $\mathbf{u}_{GF} = (0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$.

Entire frame, $\sum M_x$ ($M_x = yF_z - zF_y$):

$$\sum M_x = (1.2)(-P) - (-0.6)\frac{F_G}{\sqrt{2}} = 0 \ \Rightarrow\ P = \frac{0.6}{1.2\sqrt{2}}F_G = 0.3536\,F_G$$

With $F_G = 800$ N: $P_{\max} = (0.3536)(800) = 282.8$ N.

Member BD. FG loads BD at F with $(0, 565.7, 565.7)$ N. Let (B_x, B_y, B_z) be the force of BD on ABC; by symmetry BD feels $(-B_x, -B_y, -B_z)$ at B and $(B_x, -B_y, -B_z)$ at D:

$$\sum F_y = 565.7 - 2B_y = 0 \ \Rightarrow\ B_y = 282.8\,\text{N}, \qquad \sum F_z = 565.7 - 2B_z = 0 \ \Rightarrow\ B_z = 282.8\,\text{N}$$

B_x drops out of BD's equations, so it comes from ABC.

Member ABC. Each leg carries $P/2 = 141.4$ N at C, and the legs interact there only along x. Forces: $(0, A_y, A_z)$ at A, $(B_x, 282.8, 282.8)$ at B, $(C_x, 0, -141.4)$ at C; $\vec{r}_{AB} = (-0.15, 0.6, 0)$,

$\vec{r}_{AC} = (-0.3, 1.2, 0)$:

$$\sum F_x = B_x + C_x = 0 \;\Rightarrow\; C_x = -B_x$$
$$\sum M_{A,z} = \big[(-0.15)(282.8) - (0.6)B_x\big] - (1.2)C_x = -42.43 + 0.6B_x = 0 \;\Rightarrow\; B_x = 70.7\,\text{N}$$

Answer

$P_{\max} = 283\,\text{N}$
BD on ABC (at B): $(70.7\,\text{N},\ 283\,\text{N},\ 283\,\text{N})$
BD on EDC (at D): $(-70.7\,\text{N},\ 283\,\text{N},\ 283\,\text{N})$